

Vinh Ly

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Summary

Software Engineer with **18 years** delivering end-to-end, kernel-level services for billion-scale platforms. Owned YouTube's real-time video recommendation pipeline, Whatnot's marketplace matching engine, and Google's global search index, each built on **C++**, **Go**, **Python**, **Kubernetes** and **Linux**. Cut average request latency by 35% while supporting 2 billion daily requests, collaborating with Product, Data Science and Infrastructure partners. He has repeatedly designed fault-tolerant observability layers, mentored junior engineers, and instituted on-call rotations that reduced incident MTTR by 20%. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: C++, Go, Java, Python, SQL, TypeScript, JavaScript, HTML, CSS

Frameworks & Platforms: Kubernetes, Docker, gRPC, REST, Linux

Systems & Tools: Git, CI/CD, Docker, Prometheus, Grafana

Methodologies: Test-Driven Development, Microservices, On-call

Work Experience

Staff Software Engineer, YouTube (02/2025 - Present), Remote

- Modernized the video recommendation pipeline to process 1.2 B events per day, replacing legacy batch jobs with a streaming architecture built on **Go** and **Kubernetes**, cutting processing latency by 40%.
- Implemented a fault-tolerant observability stack using **Prometheus** and **Grafana**, enabling real-time alerting that reduced mean-time-to-detect incidents from 12 minutes to under 3 minutes.
- Led a cross-functional effort with Product and Data Science to integrate a new ranking model, delivering a 5% increase in click-through rate across core surfaces.
- Architected a kernel-level caching layer in **C++** that served 2 billion daily requests, improving cache hit ratio by 22% and saving \$3 M in compute costs.

Staff Software Engineer, Whatnot (05/2024 - 02/2025), Remote

- Collaborated with Data Engineering to launch a streaming analytics pipeline that surfaced fraud patterns, preventing \$1.2 M in fraudulent transactions within the first quarter.
- Re-engineered the marketplace matching engine in **Java** and **Kubernetes**, handling a 30% surge in concurrent users while maintaining sub-100 ms response times.
- Built a real-time inventory sync service using **Go** and **gRPC**, reducing stale listings by 45% and improving seller satisfaction scores.
- Introduced automated canary deployments via **CI/CD** pipelines, achieving zero-downtime releases for 12 major feature rollouts.

Staff Software Engineer/Manager, Google (03/2015 - 04/2024), Remote

- Directed a team of 12 engineers to redesign the global search index, using **C++** and **Python** to index 500 B documents, cutting query latency by 28%.
- Implemented a distributed tracing system with **gRPC** and **OpenTelemetry**, improving root-cause analysis speed by 35% across services.
- Scaled the ad-delivery platform to support 1.5× traffic growth, introducing autoscaling policies in **Kubernetes** that maintained 99.95% uptime.
- Mentored senior engineers on performance profiling, resulting in a 22% reduction in CPU usage for core services.

- Established on-call rotation and incident review framework, decreasing production incidents by 30% over 2 years.

Software Engineer, Google (02/2011 - 02/2015), Remote

- Created a high-throughput logging service using **Python** and **Kafka**, handling 200 M events per day with sub-50 ms latency.
- Developed core components of the search ranking pipeline in **Java**, improving relevance metrics by 12% for mobile queries.
- Optimized data ingestion pipelines, reducing ETL processing time from 8 hours to 2 hours, saving \$500 K annually.
- Collaborated with Site Reliability teams to implement health checks and automated rollbacks, decreasing service outages by 40%.

Software Design Engineer II, Microsoft (10/2007 - 12/2011), Remote

- Designed and shipped the Azure Storage client library in **C#** and **REST**, serving over 10 M developers worldwide.
- Authored performance benchmarks that identified a 25% CPU reduction opportunity, leading to a redesign of the I/O path.
- Implemented a distributed file synchronization service using **C++**, achieving sub-second consistency across data centers.
- Partnered with QA to introduce automated regression testing, cutting release verification time by 60%.

Software Design Engineer Intern, Microsoft (06/2006 - 08/2006), Remote

- Contributed to the development of a prototype compiler front-end in **C++**, adding support for new language constructs.
- Presented performance analysis findings to senior engineers, influencing a redesign that improved parsing speed by 15%.
- Wrote unit tests that increased code coverage from 45% to 78% for the assigned module.
- Documented design decisions in Confluence, creating a reference guide used by subsequent intern cohorts.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007